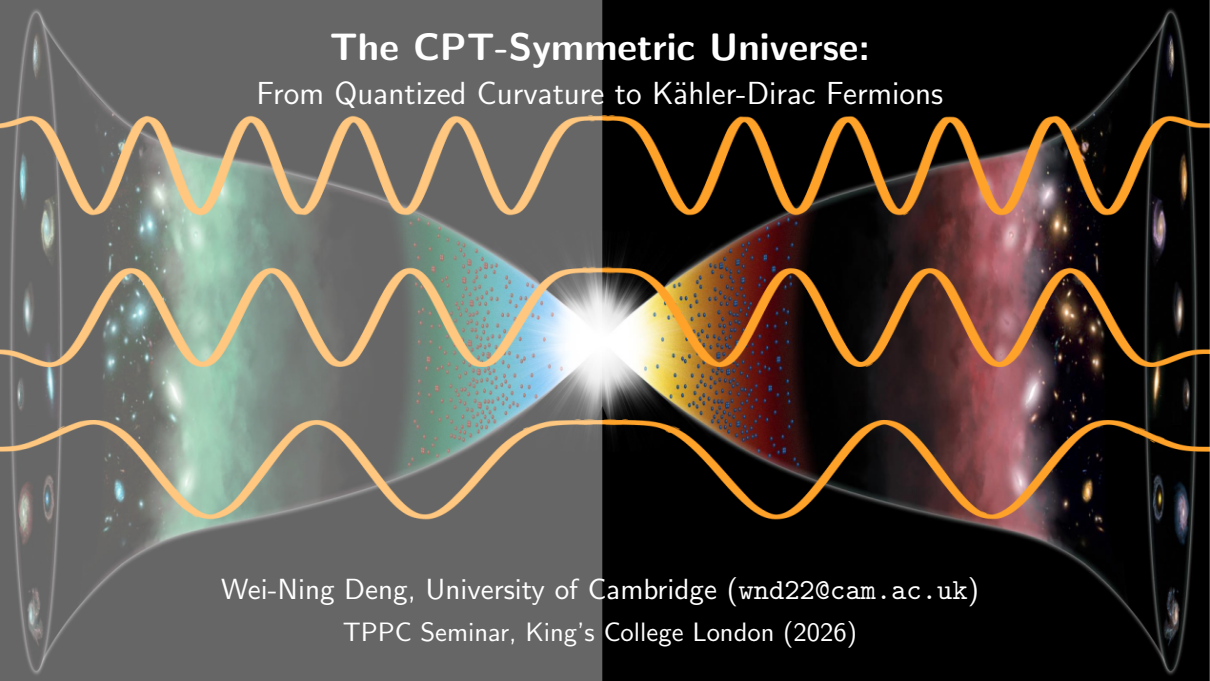


The CPT-Symmetric Universe:

From Quantized Curvature to Kähler-Dirac Fermions



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TPPC Seminar, King's College London (2026)

Outline

Part 1: Introduction – CPT-symmetric universe

Part 2: Theoretical prediction of spatial curvature (PRD 110, 103528; PRD 113, 023546)

Part 3: Kähler-Dirac fermions – Physical foundation of CPT (arXiv:2511.11548)

Future directions: Lattice KD, Composite Higgs, Quadratic gravity



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Theoretical and observational challenges

1. Problems in cosmology:

Cosmological constant problem, nature of DM, origin of matter–antimatter asymmetry

2. Problems in particle physics:

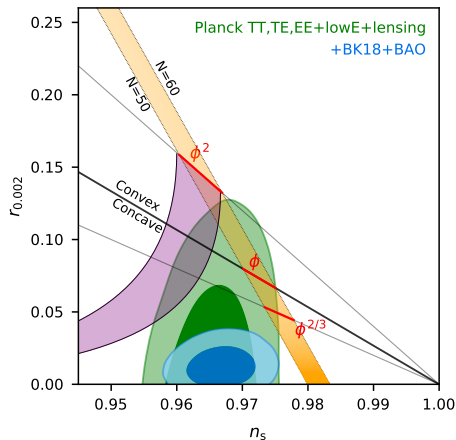
Hierarchy problem, strong CP problem, origin of 3 fermion generations

3. Observational challenges:

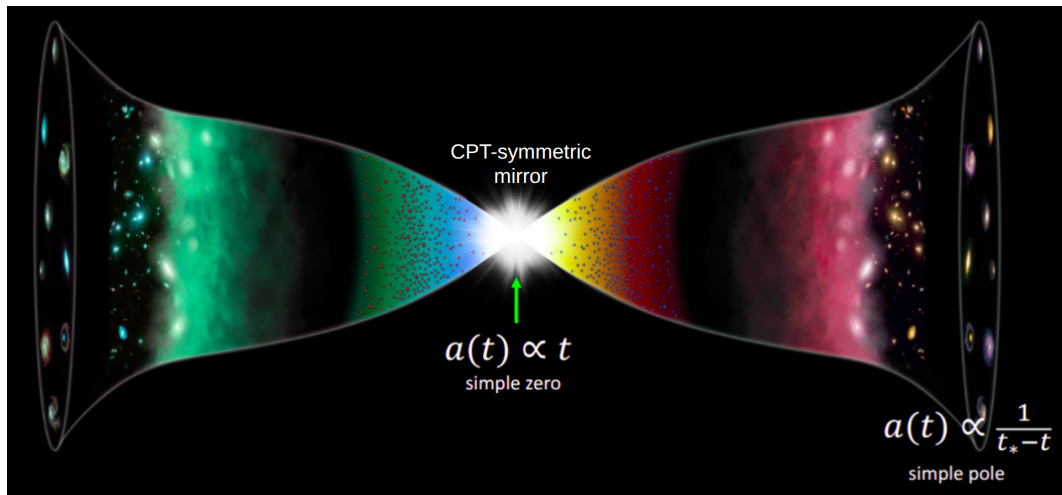
BICEP/Keck [arXiv:2203.16556]: tight constraint on inflationary B-modes ($r < 0.036$)

4. No prediction for Ω_K :

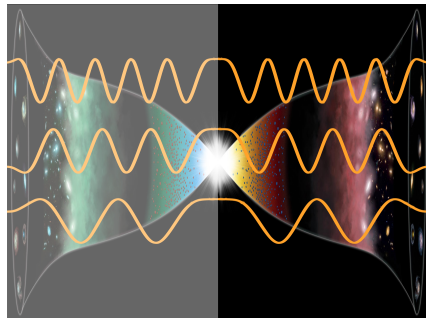
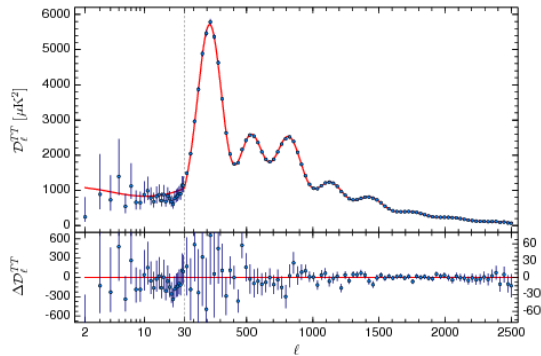
Treated as a free parameter – but Planck 2018 shows a 2σ preference for $\Omega_K < 0$



CPT-symmetric universe



CMB acoustic peaks: direct evidence for a temporal mirror



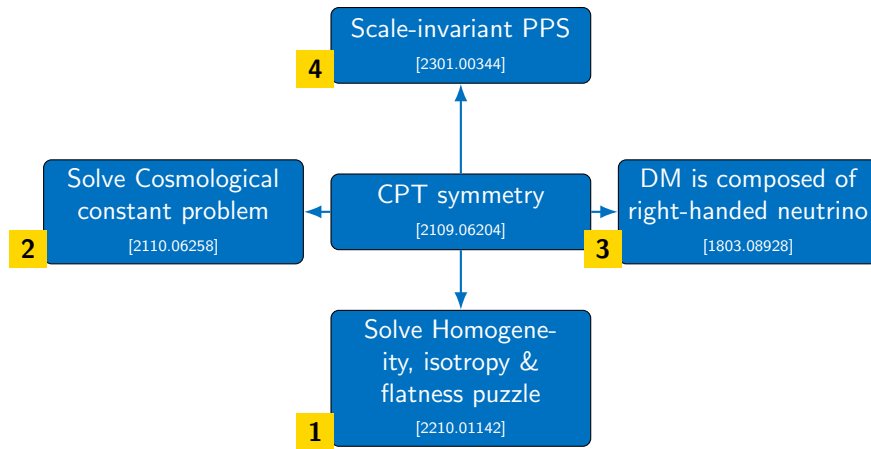
The **ringing pattern** in the CMB TT (and TE) power spectra is direct evidence that primordial density perturbations obeyed a **reflecting (Neumann) BC** at the start of the radiation era:

$$\dot{\Phi}|_{\eta=0} = 0 \quad \Longleftrightarrow \quad \text{Big Bang acts as a } \mathbf{temporal\ mirror}$$

If radiation era was *not* preceded by inflation, it's **direct obs. evidence** that BB is a temporal mirror.

CPT-symmetry universe can solve lots of problems

(Proposed by Latham Boyle and Neil Turok)



Analytic solution of Friedmann equation in conformal time

Metric:

$$ds^2 = a(t)^2 (-dt^2 + \gamma_{ij} dx^i dx^j)$$

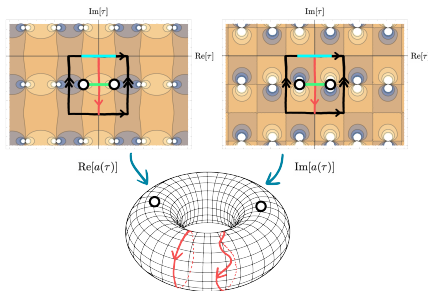
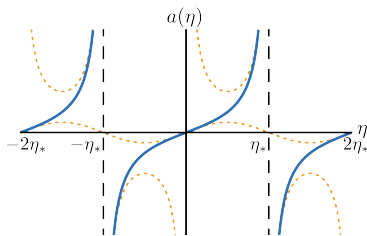
Friedmann equation (Planck units):

$$3\dot{a}^2 = \underbrace{r}_{\text{rad}} + \underbrace{\mu a}_{\text{matter}} - \underbrace{3\kappa a^2}_{\text{curvature}} + \underbrace{\lambda a^4}_{\Lambda}$$

General solution (Jacobi elliptic function):

$a(t)$ is doubly periodic in the complex t -plane.

The imaginary time period and the action computed over a period determine T_H and the gravitational entropy S_g . (thermodynamics selects the most probable universe).



Euclidean torus: imaginary-time instanton

Flat, isotropic, homogeneous universe is most probable



Gravitational UV fixed points

Problem: coupling QFT to gravity $\Rightarrow$ UV divergences in $\langle T_{\mu\nu} \rangle$, implying wildly curved spacetime on short scales.

ERGE approach: identify UV fixed points where *all* gravitational β -functions vanish.

Boyle, Turok, Vaibhav, arXiv:2509.09346

Beta functions:

$$\beta_{\Lambda/G} \propto n_0 - 2n_{1/2} + 2n_1 + 2n'_0 = 0$$

$$\beta_i (\text{four-deriv. terms}) = 0, \quad i = 1, 2, 3, 4$$

All β -functions vanish iff:

$$n_{1/2} = 4n_1, \quad n'_0 = 3n_1, \quad n_0 = 0$$

Conformally coupled fields:

- n_1 : gauge fields
- $n_{1/2}$: Weyl/Majorana fermions
- n_0 : Klein-Gordon scalars
standard 2-derivative
- n'_0 : Fradkin-Tseytlin (FT) scalars
4-derivative conformally coupled

At the fixed point:

- $G(k) \sim k^{-2}$ in UV (“softening”)
- $\langle T_{\mu\nu} \rangle$ and all $\langle TT \rangle$ UV-finite
- **Cosmological constant problem solved**

Standard Model at the gravitational fixed point

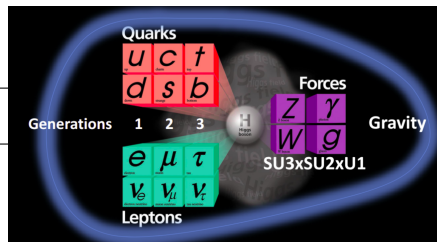
SM gauge group $SU(3) \times SU(2) \times U(1)$:

$n_1 = 8 + 3 + 1 = 12$ gauge fields

	Required	SM (incl. ν_R)
Fermions $n_{1/2}$	$4 \times 12 = 48$	$3 \times 16 = 48 \checkmark$
KG scalars n_0	0	Higgs $\rightarrow$ composite
FT scalars n'_0	$3 \times 12 = 36$	0 (to predict!)

Predictions:

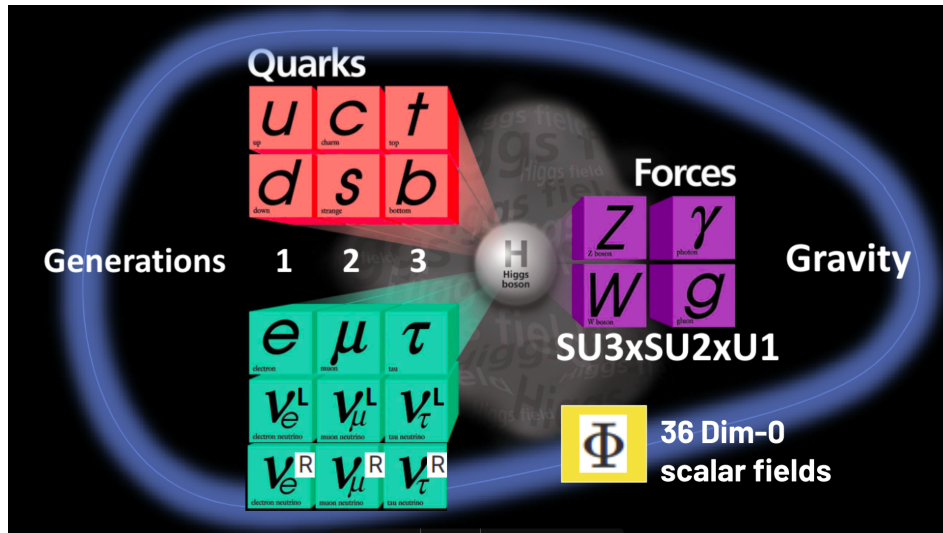
1. **3 generations + ν_R :** $n_{1/2} = 48$ requires right-handed neutrinos
2. **Composite Higgs:** $n_0 = 0$ forbids fundamental KG scalars
3. **36 FT (dim-0) scalars** also generate scale-invariant primordial spectrum!



IR: positive G + arbitrarily small Λ

arXiv:2509.09346

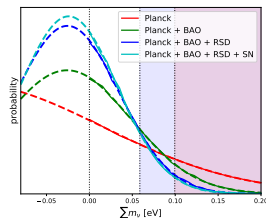
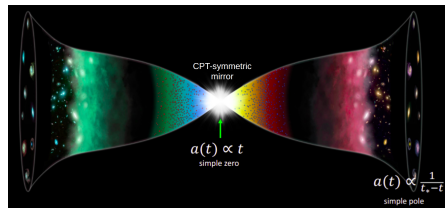
Extended Standard Model: $3 \nu_R + 36 \text{ dim-0 scalars}$



Right-handed neutrino as dark matter

CPT-symmetric universe predicts:

- $\mathbb{Z}_2$ symmetry stabilises *one* $\nu_R \Rightarrow$
lightest neutrino is massless
- The stable ν_R is produced via metric time-dependence when $H \sim M$
(analogous to Hawking radiation)
- ν_R dark matter mass:
 $M_{\nu_R} \approx 5 \times 10^8 \text{ GeV}$ [arXiv: 1803.08928]
- **Falsifiable:** lightest ν massless $\Rightarrow$
normal hierarchy, $\sum m_\nu \approx 0.06 \text{ eV}$
EUCLID, CMB-S4, LSST will test this



sum neutrino mass (eBOSS 2007.08991)

Dim-0 scalar field seeds scale-invariant PPS

$$S = -\frac{1}{2} \int d^4x \sqrt{-g} \varphi \Delta_4 \varphi$$

$$\langle \delta\varphi_i(\eta, \mathbf{x}) \delta\varphi_j(\eta, \mathbf{x}') \rangle = \delta_{ij} \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}}{4ak^3}$$

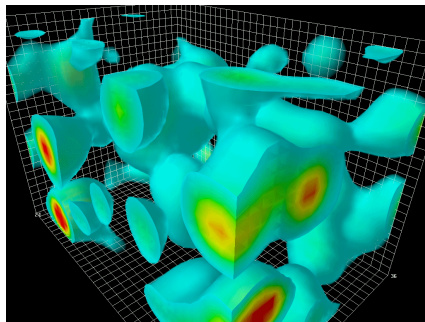
$$\mathcal{P}_{\mathcal{R}} = \frac{3^2 5^3}{7(2\pi)^4} \left(\frac{c_{\beta}^{SM}}{\mathcal{N}_{eff}} \right)^2$$

Predicted: $A = 12.9 \pm 4.5 \times 10^{-10}$,

$$n_s \approx 1 - 7\alpha_3(M_P)/\pi = 0.958$$

Observed (Planck): $A = 21 \pm 0.3 \times 10^{-10}$,

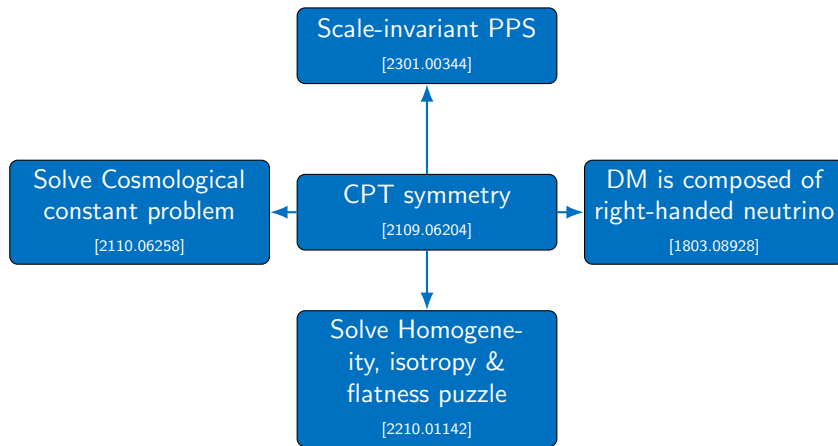
$$n_s = 0.9587 \pm 0.0056$$



Quantum fluctuations of dim-0
scalars seed scale-invariant PPS

CPT-symmetry universe can solve lots of problems

(Proposed by Latham Boyle and Neil Turok)



How about perturbations?

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Perturbations must be CPT-symmetric: the resonant cavity

Perturbations oscillate in conformal time

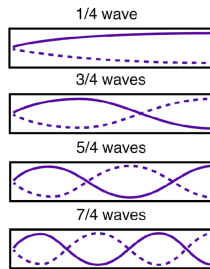
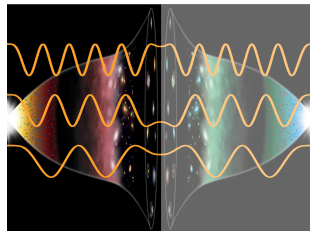
$$\Phi(k, \eta) \propto \cos \theta(k, \eta)$$

Consequences for perturbations:

- Perturbations must be (anti-)symmetric, otherwise would blow up at the future Big Bang (anti-universe). Lasenby et al. [arXiv: 2104.02521].
- This is the resonant cavity condition: only “standing-wave” modes are allowed
- Analogy: a resonant cavity selects discrete frequencies

$$\theta(k, \eta = \eta_{\infty}) = m \frac{\pi}{2}, \quad m \in \mathbb{N}$$

⇒ Only a **discrete set of k** is allowed.



Spatial boundary condition in a closed universe

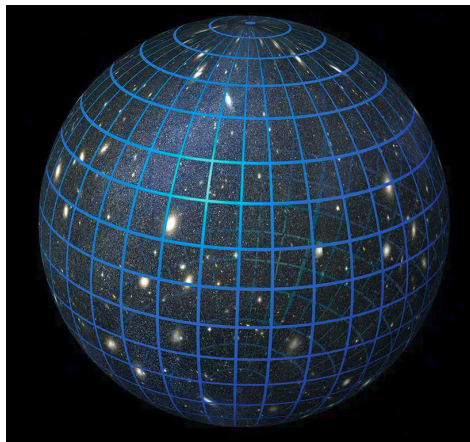
Closed universes have finite space $\Rightarrow$ an additional spatial boundary condition (in spherical harmonics)

$\Rightarrow$ wave vectors have to be integers:

$$k = n \in [1, 2, 3, \dots]$$

Matching the two discrete k sets

$\Rightarrow$ **constrains the curvature value.**

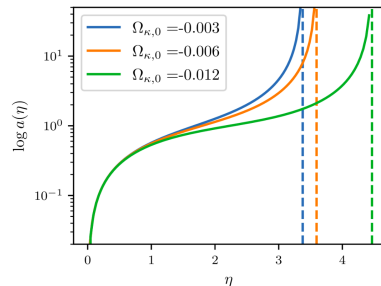


Closed universe: finite spacetime volume

Curvature controls the spacing – matching the two discrete sets

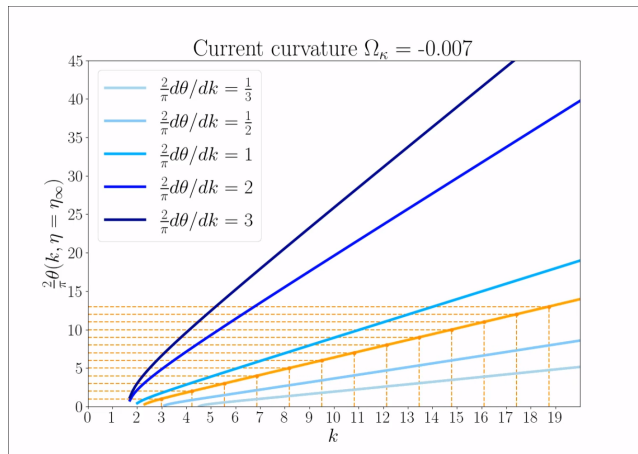
Larger $|\Omega_K| \rightarrow$ larger η_∞

Since $\Delta k \propto 1/\eta_\infty$, larger curvature means **smaller** Δk .



$\Omega_{K,0}$ more negative = universe lives longer

Idea: Adjust Ω_K until the temporal discrete- k values become integers (required by the spatial BC).



Result: predicted Ω_K and observational status

Result (toy model, radiation only):

- The integer- k condition selects discrete values of $\Omega_K < 0$.
- Allowed gradient diverges at $\Omega_{K,0} \simeq -0.012$ (big crunch limit).

Observational status

PRD 110, 103528

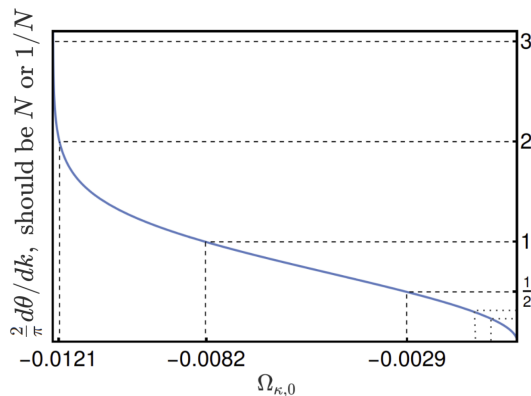
Planck 2018 + lensing:

$$\Omega_K = -0.044^{+0.018}_{-0.015}$$

(2σ preference for closed universe)

Handley reanalysis:

$\sim 3\sigma$ Bayesian tension for $\Omega_K < 0$

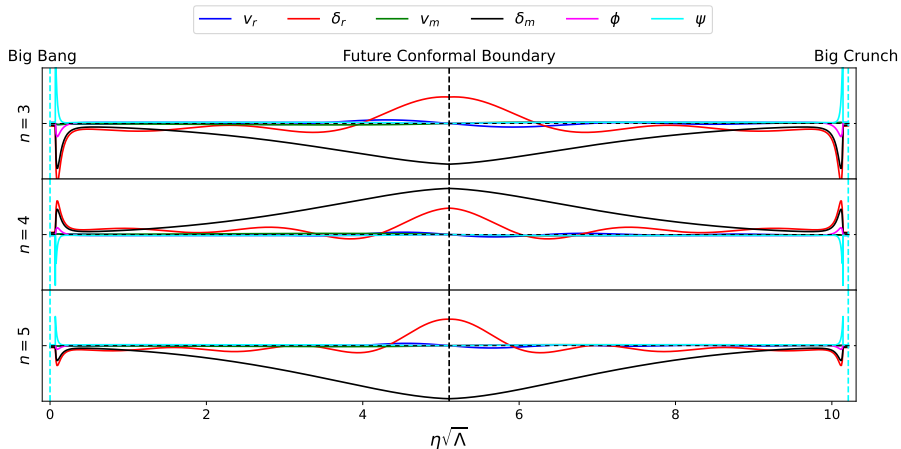


Δk (spacing of allowed k) vs Ω_K .

Solve Perturbations in real cosmology

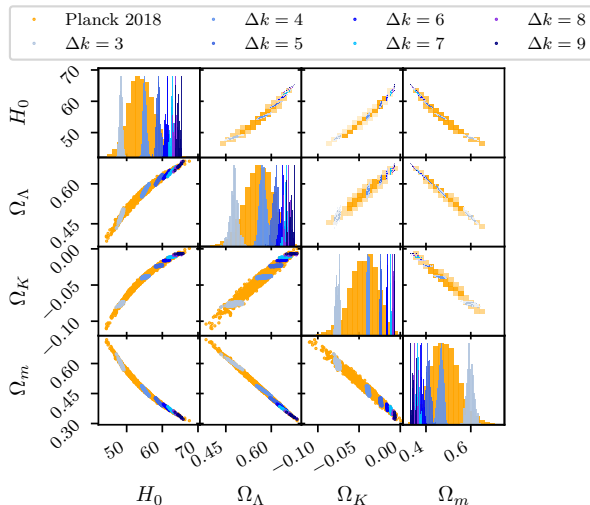
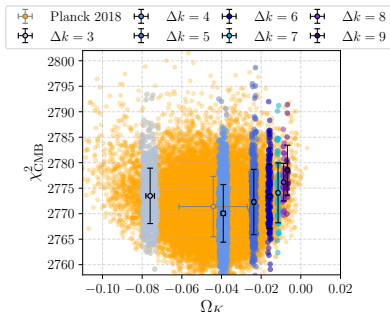
First few allowed modes of Perturbations

$\Phi(k, \eta) \propto \cos \theta(k, \eta)$, Φ : Newtonian potential, V : velocity, δ : density perturbation



Result: Planck MCMC triangle plot (Real Cosmology)

- Calculate $\Delta k \in \mathbb{N}$ numerically with radiation anisotropy and higher order Boltzmann terms.
- Filter Planck MCMC data with the $\Delta k \in \mathbb{N}$ constraint.
- Need joint analysis to break the degeneracy.

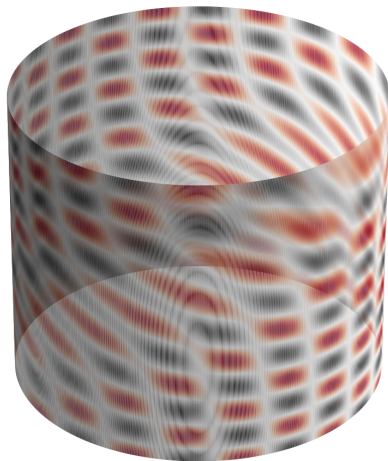


Low- k eigenmodes: mathematical consistency

1. Large k : allowed wave vectors equally spaced (handled).
2. Low k : valid solutions must satisfy *both* spatial *and* temporal BCs simultaneously
 $\Rightarrow$ linear combination of both bases:

$$\Phi(x, t) = \sum_n c_n \phi_n = \sum_m \tilde{c}_m \tilde{\phi}_m$$

3. Coefficients = eigenvectors of mixing matrix M with eigenvalue = 1.



Eigenvalue = 1 selects valid low- k modes.

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Fermion doubling problem on the lattice

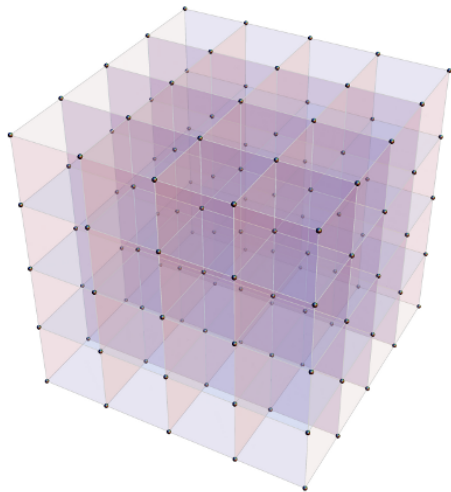
Wilson's lattice gauge theory works

beautifully for bosons:

- 0-forms (scalars) $\rightarrow$ vertices
- 1-forms (gauge fields) $\rightarrow$ links
- 2-forms (field strengths) $\rightarrow$ plaquettes

Problem with spinors: no natural p -form interpretation $\Rightarrow$ placed on vertices $\Rightarrow$ **extra doublers in the continuum limit**

Chiral theory $\xrightarrow{\text{lattice}}$ non-chiral theory
(Nielsen–Ninomiya, 1981)



Kähler-Dirac (KD) fermions

Key idea: KD fermions are p -forms – they live naturally on any cell complex.

KD field Φ is a *polyform*: sum of p -forms ($p = 0, \dots, 4$) with 16 complex components.

KD action and equation:

$$S_{KD} = \langle \bar{\Phi} | (K - m) \Phi \rangle, \quad (d - d^\dagger - m) \Phi = 0$$

where $K = d - d^\dagger$, $-K^2 = \square$ (wave operator on forms).

Replacing $dx^\mu \rightarrow \gamma^\mu$ assembles the 16 components into a 4×4 matrix Ψ :

$$(i\gamma^\mu \partial_\mu - m)\Psi = 0 \quad \Rightarrow \quad S_{KD} = \int d^4x \operatorname{Tr}[\bar{\Psi}(i\not{\partial} - m)\Psi]$$

with $\bar{\Psi} \equiv \gamma^0 \Psi^\dagger \gamma^0$ (extra γ^0 on the left!).

Symmetry: $\Psi' = U_\Lambda \Psi V$, where $(U_\Lambda, V) \in \operatorname{Spin}(3, 1) \times U(2, 2)$. When restrict $V = U_\Lambda^{-1} \in SU(2, 2)$, behave like forms.

The wrong-sign problem in Lorentzian signature

Diagonalise the extra γ^0 : $\gamma^0 = \Omega \Lambda \Omega^T$, $\Lambda = \text{diag}(+1, +1, -1, -1)$.

In terms of $\Psi' \equiv \Psi \Omega$:

$$S_{KD} = \int d^4x \text{Tr} \left[\Lambda \Psi'^{\dagger} \gamma^0 (i \not{\partial} - m) \Psi' \right]$$

- Columns 1–2 of Ψ' : **right sign** $\Rightarrow$ positive norm
- Columns 3–4 of Ψ' : **wrong sign** $\Rightarrow$ **negative norm states (ghosts)!**

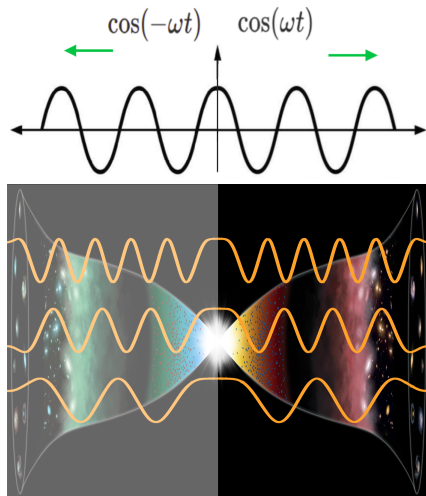
Resolution (Donoghue & Menezes) : a wrong-sign action = reversal of causality arrow = ***PT* reversal** [1908.04170].

$\Rightarrow$ the two pairs of columns live on **two sheets** of spacetime, related by *PT* symmetry (or, equivalently, $i \leftrightarrow -i$).

Two-sheeted spacetime

- **Forward sheet** ($\Lambda_{ii} = +1$, cols 1–2): standard quantum mechanics with e^{iS} , $[x, p] = +i$
- **Backward sheet** ($\Lambda_{ii} = -1$, cols 3–4): uses e^{-iS} , $[x, p] = -i$
- The two sheets are *PT* **mirrors** of each other
- Positive energy, positive norm on *both* sheets

“Our” Big Bang is the junction of the two sheets – exactly the CPT-symmetric universe picture!



Physical explanation of the movie Tenet?



KD-Majorana condition

Problem: Two independent non-interacting sheets would double the particle content.

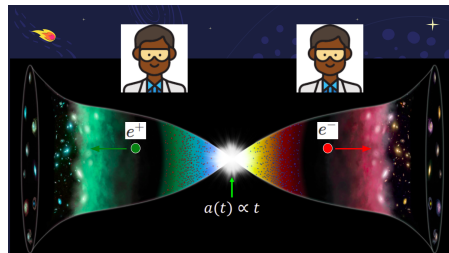
Solution: Impose the **KD-Majorana condition**

$$\Psi^c \equiv (i\gamma^2)\Psi^*(i\gamma^2) = \Psi$$

In terms of Ψ' , the four columns are forced to take the form:

$$\Psi' = (\psi_1, \psi_2, -\psi_2^c, \psi_1^c)$$

- A LH particle on the forward sheet is *always* paired with a RH anti-particle on the backward sheet
- **Physical meaning:** CPT symmetry – every particle has a mirror (anti-)partner on the other sheet
- No unphysical doubling of the SM



Relation to the Standard Model

One SM generation (including ν_R) = 16 Weyl spinors = 4 KD-Majorana fields.

Following Pati-Salam: allocate to four copies $\Psi'_{A=0,1,2,3}$ ([S. Catterall, 2209.03828])

$$\Psi'_0 = \chi_0 + \chi_0^c, \quad \chi_0 = \begin{pmatrix} \nu_L & e_L \\ \nu_R & e_R \end{pmatrix} \quad (\text{leptons})$$

$$\Psi'_i = \chi_i + \chi_i^c, \quad \chi_i = \begin{pmatrix} u_L^i & d_L^i \\ u_R^i & d_R^i \end{pmatrix} \quad (\text{quarks, color } i = 1, 2, 3)$$

Fermion kinetic terms unify:

$$\mathcal{L}_{\text{KD}} = \text{Tr} [\bar{\Psi}_A (i \not{\partial}) \Psi_A], \quad A = 0, \dots, 3$$

Gauging $SU(3) \subset SU(4)$ and $SU(2) \times U(1) \subset U(2, 2)$ (the internal symmetry) reproduces *all* SM fermion kinetic terms.

Yukawa couplings unify in KD formalism

Standard Yukawa Lagrangian (4 separate terms):

$$\tilde{h}^T \bar{q}_L u_R y_u + h^T \bar{q}_L d_R y_d + \tilde{h}^T \bar{\ell}_L \nu_R y_\nu + h^T \bar{\ell}_L e_R y_e + \text{h.c.}$$

KD formulation (single compact expression):

$$\mathcal{L}_Y = \text{Tr} \left[\mathcal{H}^T \widehat{\Psi}_{A,L}'^i \Psi_{A,R}'^j \mathcal{Y}_A^{ij} \right] + \text{h.c.}$$

where $\mathcal{H}$ is the *KD Higgs field* and $\mathcal{Y}_A^{ij}$ the *KD Yukawa matrix*.

- Expanding for leptons reproduces exactly the two lepton Yukawa terms
- Forward sheet $\leftrightarrow$ particles; backward sheet $\leftrightarrow$ antiparticles
- All four SM Yukawa sectors unified in one line

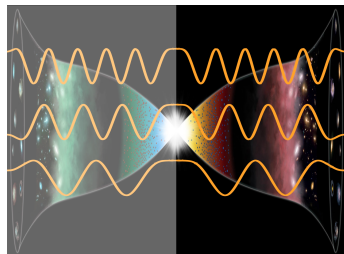
Connection to the CPT-symmetric universe

Question: *Why* does the Big Bang act as a CPT mirror?

The CPT-symmetric universe was motivated by observations (explain CMB and lots of cosmological phenomena).

KD answer: Two sheeted is required for solving negative norm problem. The KD-Majorana condition *requires* every particle to have a mirror (anti-)partner on the other sheet.

The Big Bang is precisely the junction where the two spacetime sheets meet.



$$\underbrace{\text{KD-Majorana}}_{\text{field theory}} \longleftrightarrow \underbrace{\text{CPT mirror at bang}}_{\text{cosmology}}$$

Black mirror: an alternative black holes theory

Standard picture: black hole has an interior.

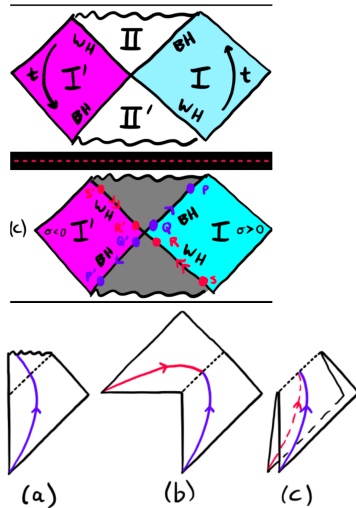
- Problem: Hidden from Observation? Curvature singularities? Information paradox? Violates all global symmetries (CPT, B-L)?

Black mirror proposal [Tzanavaris, Latham
[arXiv: 2412.09558]]

- The interior *does not exist*
- The horizon mirrors our exterior to the “other” exterior (anti-universe)
- Infalling particles *annihilate* with anti-partners from the anti-universe at the horizon

KD-Majorana condition: explains why every infalling particle is paired with a mirror anti-partner.

Open question (for London): does the theory modify quasi-normal modes?



Theoretical description of Tenet?!

Same as CPT-symmetric universe:

1. There is *you* and *anti-you* in the universe and anti-universe
2. Time goes backward in the anti-universe
3. The doors connecting universe and anti-universe are **black holes**

Difference:

1. You and anti-you **annihilate** at the transition door
2. If viewed from our universe: you come out of a **white hole** and find that time goes backward. However, you *cannot change anything* – only re-experience the past



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Future direction 1: KD fermions on the lattice

Boyle, Vaibhav (in preparation)

Problem: chirality in Euclidean signature

KD operator K anti-commutes with $\Gamma \Rightarrow$ Euclidean action $\langle \bar{\Psi}, K\Psi \rangle$ mixes chiralities: *cannot write a purely chiral action*

Solution: mirror fermion space (Barrett [arXiv: 2403.18428])

Double the Hilbert space $\mathcal{H}' = \mathcal{H} \oplus \mathcal{H}$, with charge conjugation J interchanging the two copies: $S_{\text{chiral}} = \frac{1}{2} \langle J'\Psi_L, K'\Psi_L \rangle$

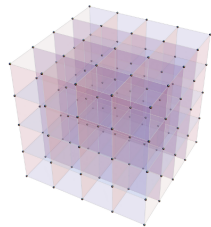
't Hooft anomaly $\Rightarrow$ Pati-Salam

Cancelling the $\mathbb{Z}_4$ chiral anomaly on the lattice requires **4 copies** of reduced KD fermions $\Rightarrow$ gauge group $SU(4) \times SU(2)_L \times SU(2)_R$

Two-sheeted spacetime is not just cosmology!

$\mathcal{H}' = \mathcal{H} \oplus \mathcal{H} \equiv \text{universe} + \text{anti-universe}$

Required to simulate a **chiral gauge theory** on the lattice



Nielsen-Ninomiya
avoided:
equal $n_L = n_R$ in
Pati-Salam

Chiral anomaly
preserved
on the lattice

Audience:
Alexandre, Sarkar

Future direction 2: composite Higgs from dim-0 scalars

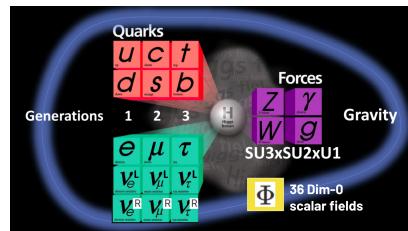
Recall: anomaly cancellation predicts $n_{s,1} = 0$ –
no fundamental dimension-1 scalars.

⇒ **The Higgs must be composite.**

The challenge:

- Build a viable composite Higgs model from the collective dynamics of the **36 dim-0 scalar fields**
- Constrained by LHC data: no new physics at the TeV scale (κ_V, κ_F within 5% of SM)
- Must reproduce Higgs mass, couplings, and EW symmetry breaking without a fine-tuned hierarchy

Getting this right would unify the origin of cosmic structure and mass in a single framework.



Audience: Ellis, Tevong You (Higgs physics, SM EFT, composite Higgs)

Future direction 3: quadratic gravity as architect of the PPS

A deeper possibility: the scale-invariant power spectrum may not require any new matter fields at all.

Quadratic gravity (Stelle, 1977):

- Renormalizable theory of quantum gravity:
$$\mathcal{L} = \frac{2}{\kappa^2} R + \frac{1}{6f_0^2} R^2 - \frac{1}{2\xi^2} C_{\mu\nu\alpha\beta} C^{\mu\nu\alpha\beta}$$
- The ghost problem in quadratic gravity $\Rightarrow$ resolved by the Donoghue–Menezes PT reversal [arXiv:2112.01974].
- Initial results (Boyle): quadratic gravity at the Big Bang can *mimic* the dim-0 scalar effect $\Rightarrow$ scale-invariant PPS without any new matter fields!
- Problem: need new explanation for cosmological constant problem.

Problem: dim-0 scalars vs. quadratic gravity. Which one is the true architect of the scale-invariant PPS?

Audience: Mavromatos, Sakellariadou, Lim, Gregory (quantum gravity, spectral geometry)

Summary

Part 1: Introduction

- CPT-symmetric universe solves flatness, CC, DM, PPS
- Extended SM: $3 \nu_R + 36$ dim-0 scalars; Higgs composite
- ν_R DM ($M \approx 5 \times 10^8$ GeV); lightest ν massless

Part 2: Quantized curvature

- CPT $\Rightarrow$ perturbations must be (anti-)symmetric
- Temporal + spatial BCs $\Rightarrow$ discrete k
- Predicts $\Omega_K < 0$; 2σ Planck preference

Future directions

- KD on lattice $\Rightarrow$ chiral SM (lattice consistency)
- Composite Higgs from 36 dim-0 scalars (LHC test)
- Quadratic gravity as origin of scale-invariant PPS

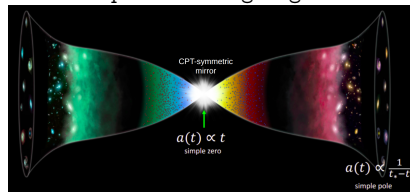
Part 3: Kähler-Dirac fermions arXiv:2511.11548

- KD fermions = polyforms; live naturally on the lattice
- Wrong-sign columns $\Rightarrow$ two-sheeted spacetime (PT mirrors; links to Alexandre et al.)
- KD-Majorana: every particle paired with mirror anti-partner
- Unifies all SM fermion kinetic + Yukawa terms
- Explains *why the Big Bang is a CPT mirror*
- ν_R required by geometry

Thank you!

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Backup: Two approaches to CPT at the quantum gravity interface

Both probe CPT in regimes where quantum gravity effects become relevant.

CPT-symmetric universe

Boyle–Turok

- Universe + anti-universe pair, joined at the Big Bang
- CPT is an **exact** symmetry of the full spacetime
- Predictions: ν_R as DM, massless lightest ν , scale-invariant PPS

CPT violation from spacetime foam

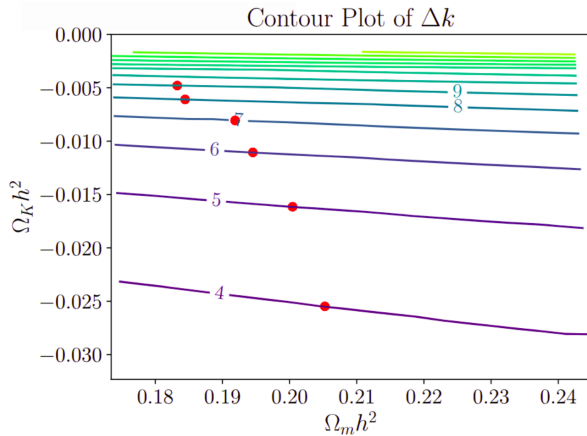
Mavromatos et al.

- Planck-scale foam induces **effective CPT violation**
- Quantum decoherence of propagating matter
- Predictions: modified EPR correlations in neutral mesons, neutrino anomalies

	CPT-symmetric universe	CPT foam	Standard inflation
CPT status	exact	violated	unspecified
Quantum gravity	none	considered	depends (ex: string)
DM candidate	ν_R	—	various (ex: axions)
Testable now?	yes	yes	yes

Backup: contour plot of Δk

$$c(\Omega_K, \Omega_m, \Omega_b, h) = N \in \mathbb{N}$$



Backup: CMB power spectrum comparison

Modify Boltzmann code CLASS to see how sparse wave vectors affect CMB power spectrum.

1. More fluctuations if N and curvature ($\tilde{\kappa}$) become large.
2. However, since $\Delta\tilde{k} = N\sqrt{\tilde{\kappa}}$, N and $\tilde{\kappa}$ are negatively correlated $\rightarrow$ never both large.
3. The fluctuation behaviour could be ignored.

